① LU für

a) $A = \begin{pmatrix} 1 & 3 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & -3 & 0 \\ 1 & 7 & 4 & 1 \end{pmatrix}$

b) $A = \begin{pmatrix} 0 & 0 & 3 & 1 \\ 2 & 1 & 2 & 0 \\ 8 & 8 & 0 & 0 \\ 4 & 6 & 2 & 4 \end{pmatrix}$

$\Big\}$ L, U, P

a)

$$\begin{array}{c|ccc} 1 & 3 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & 1 & -3 & 0 \\ 1 & 7 & 4 & 1 \end{array}$$

$p_1 = 1$

l_{i1}

$$\begin{array}{c|c|cc} 1 & 3 & 0 & 1 \\ \hline 0 & 1 & 1 & 2 \\ 2 & -5 & -3 & -2 \\ 1 & 4 & 4 & 0 \end{array}$$

$p_2 = 2$

$$\begin{array}{cc|cc} 1 & 3 & 0 & 1 \\ 0 & 1 & 1 & 2 \\ 2 & -5 & 2 & 8 \\ 1 & 4 & 0 & -8 \end{array}$$

$p_3 = 3$

$$\begin{array}{cccc}
1 & 3 & 0 & 1 \\
0 & 1 & 1 & 2 \\
2 & -5 & 2 & 8 \\
1 & 4 & 0 & -8
\end{array}$$

$$L = \begin{pmatrix} 1 & & & 0 \\ 0 & 1 & & \\ 2 & -5 & 1 & \\ 1 & 4 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} 1 & 3 & 0 & 1 \\ & 1 & 1 & 2 \\ 0 & & 2 & 8 \\ & & & -8 \end{pmatrix}$$

$$p = (1, 2, 3)$$

b)

$$\begin{array}{cccc}
0 & 0 & 3 & 1 \\
2 & 1 & 2 & 0 \\
8 & 8 & 0 & 0 \\
4 & 6 & 2 & 4
\end{array} \qquad \underline{p_1 = 2}$$

$$\begin{array}{cccc}
2 & 1 & 2 & 0 \\
\hline
0 & 0 & 3 & 1 \\
8 & 8 & 0 & 0 \\
4 & 6 & 2 & 4
\end{array}$$

$$
\begin{array}{c|ccc}
2 & 1 & 2 & 0 \\
\hline
0 & 0 & 3 & 1 \\
4 & 4 & -8 & 0 \\
2 & 4 & -2 & 4
\end{array}
\qquad p_2 = 3
$$

$$
\begin{array}{c|ccc}
2 & 1 & 2 & 0 \\
\hline
4 & 4 & -8 & 0 \\
0 & 0 & 3 & 1 \\
2 & 4 & -2 & 4
\end{array}
$$

$$
\begin{array}{cc|cc}
2 & 1 & 2 & 0 \\
\hline
4 & 4 & -8 & 0 \\
0 & 0 & 3 & 1 \\
2 & 1 & 6 & 4
\end{array}
\qquad p_3 = 3
$$

$$
\begin{array}{cccc}
2 & 1 & 2 & 0 \\
4 & 4 & -8 & 0 \\
0 & 0 & 3 & 1 \\
2 & 1 & 2 & 2
\end{array}
$$

$$L = \begin{pmatrix} 1 & & & \\ 4 & 1 & 0 & \\ 0 & 0 & 1 & \\ 2 & 1 & 2 & 1 \end{pmatrix} \qquad U = \begin{pmatrix} 2 & 1 & 2 & 0 \\ & 4 & -8 & 0 \\ 0 & & 3 & 1 \\ & & & 2 \end{pmatrix}$$

$$p = (2, 3, 3)$$